Reading a research claim

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

Ask which variable, denominator and comparison the claim compresses.

Teach faithful claim reading and missing uncertainty without accepting or dismissing effects beyond evidence.

Keep one habit: Model main rates → contrast doubling and point changes → inspect paired ambiguities → assess universal-speed rewrite → retain missing methods/evidence.

MAIN LESSON · SECTION 1

Prepare the connected research investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of reading a research **claim**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Run the investigation, with workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from A question that **evidence** can answer. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of a question that **evidence** can answer

Reference example, not the only acceptable wording: Teacher explanation: A research question identifies an outcome, comparison and conditions that **evidence** could answer. A broad aspiration needs narrowing before a study.

For fictional classroom retrieval, ask whether method B returns more relevant records than lexical baseline A on a reserved judged query set at cutoff3. Define eligible corpus and relevance rules before running.

Pictured end goal

Explain reading a research **claim** using the records and boundaries of this fictional investigation.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Teach the learning cycle

Retrieval answer guidance

A question that evidence can answer

Compact alternative: two 45-minute sessions for the core explanation and W1–W5 only. The main plan uses three sessions; schedule further time for the complete investigation, project, visual journey and any code examples. Reading a **claim** requires its method, data, comparison and uncertainty. Titles and abstracts may omit conditions crucial to interpretation.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.
</details>

Outcomes

Explain the **mechanism** and terminology of reading a research **claim**

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach denominator, outcome, comparison and uncertainty as one reading task

> Keep main eight-case supported counts distinct from picture ten-case correctness.

Definitions and complete fictional claim contract

Reading a **claim** requires its method, data, comparison and uncertainty. Titles and abstracts may omit conditions crucial to interpretation.

Separate the **claim** from supplied **evidence**. Record dataset, unit, baseline, metric and limitations. Ask whether the statement extends beyond those **observation**s.

Claim-audit item	Complete supplied fictional evidence
Main dataset/unit	Eight fixed constructed answer cases; case is the reported unit
Main baseline A	5 supported answers / 8 cases
Main method B	6 supported answers / same 8 cases

Claim-audit item	Complete supplied fictional evidence
Main outcome	Supplied supported-answer flag; do not relabel it learner improvement or overall usefulness
Main descriptive change	62.5% to 75%; +12.5 percentage points; ratio 1.2
Missing method details	Exact support rubric, versions, evaluation conditions and sampling/uncertainty analysis not supplied by the headline
Separate picture	Correct-answer count 2/10 to 4/10 on same ten cases with fixed scoring
Picture descriptive change	20% to 40%; +20 percentage points; rate ratio 2, relative increase 100%
Limits	No population interval, significance result, causal effect or individual-learning evidence supplied
Evidence status	All counts and any added possible paired records are fictional examples, not a real paper or new experiment

Main arithmetic ledger and checked reasoning

Main fictional comparison	Baseline A	Method B	Supported descriptive statement
Supported count / same cases	5 / 8	6 / 8	One more supported answer in these counts
Observed percentage	62.5%	75%	+12.5 percentage points
Rate ratio	Reference	(6/8) / (5/8) = 1.2	20% relative increase in this measured rate
Case-level transitions	Not supplied	Not supplied	Totals do not show who gained or regressed
Broader inference	No sampling/uncertainty analysis	No general task or learner outcomes	Do not claim universal superiority or causal effect

Main 5/8 = 62.5%, 6/8 = 75%, difference 12.5 percentage points, ratio 1.2. Outcome is supported answers, not learner improvement. Picture separate 2/10 = 20%, 4/10 = 40%, ratio 2 and 100% relative increase, absolute +20 points; six new-version cases still incorrect. No individual learner data or significance/causal result supplied.

Two added possible paired patterns share main totals but differ in gains/losses; do not call them the paper's actual case records. Selected four-fixture medians 2 versus 3 seconds give an observed one-second median difference locally; record workload, environment, quality and timing boundaries, without universal speed or unprovided raw timing claims.

Reading a research claim — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
A fictional report says method B is better after 6/8 supported answers versus baseline 5/8 on the same constructed cases.	Separate the claim from supplied evidence. Record dataset, unit, baseline, metric and limitations. Ask whether the statement extends beyond those observations.	B has 75% versus 62.5%, a 12.5-percentage-point difference on eight cases. Better in this measured outcome does not establish broader tasks or uncertainty.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: B has 75% versus 62.5%, a 12.5-percentage-point difference on eight cases. Better in this measured outcome does not establish broader tasks or uncertainty. Qualification: A numerical improvement can be real in the sample while the broader **claim** remains unsupported. Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a **claim** resting on unmeasured **evidence**.

A fictional report says method B is better after 6/8 supported answers versus baseline 5/8 on the same constructed cases.

B has 75% versus 62.5%, a 12.5-percentage-point difference on eight cases. Better in this measured outcome does not establish broader tasks or uncertainty.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Unit, relative-change and inference probes

Claim-audit probe	Expected interpretation
Count improves with same denominator	Compute a descriptive rate difference under that outcome
“Better” omits metric or baseline	Ask what was measured and compared
Percent change confused with percentage points	Write both starting and ending rates before calculating
Rate ratio with zero baseline	Undefined; do not invent a finite doubling ratio
Same aggregate counts, different paired records	Individual gains/losses are not identified by totals
Headline names learners but table counts answers	Unit/outcome mismatch
No uncertainty analysis supplied	No significance or population-precision claim established
Matched cases alone	Useful comparability condition, not automatic causal identification
One small fictional report	No universal task, learner or deployment conclusion

Separate prompt comparison and full absolute/relative arithmetic

Read a **claim** with its denominator. A fictional paper headline says a prompt is twice as effective. Its table reports four correct cases instead of two.

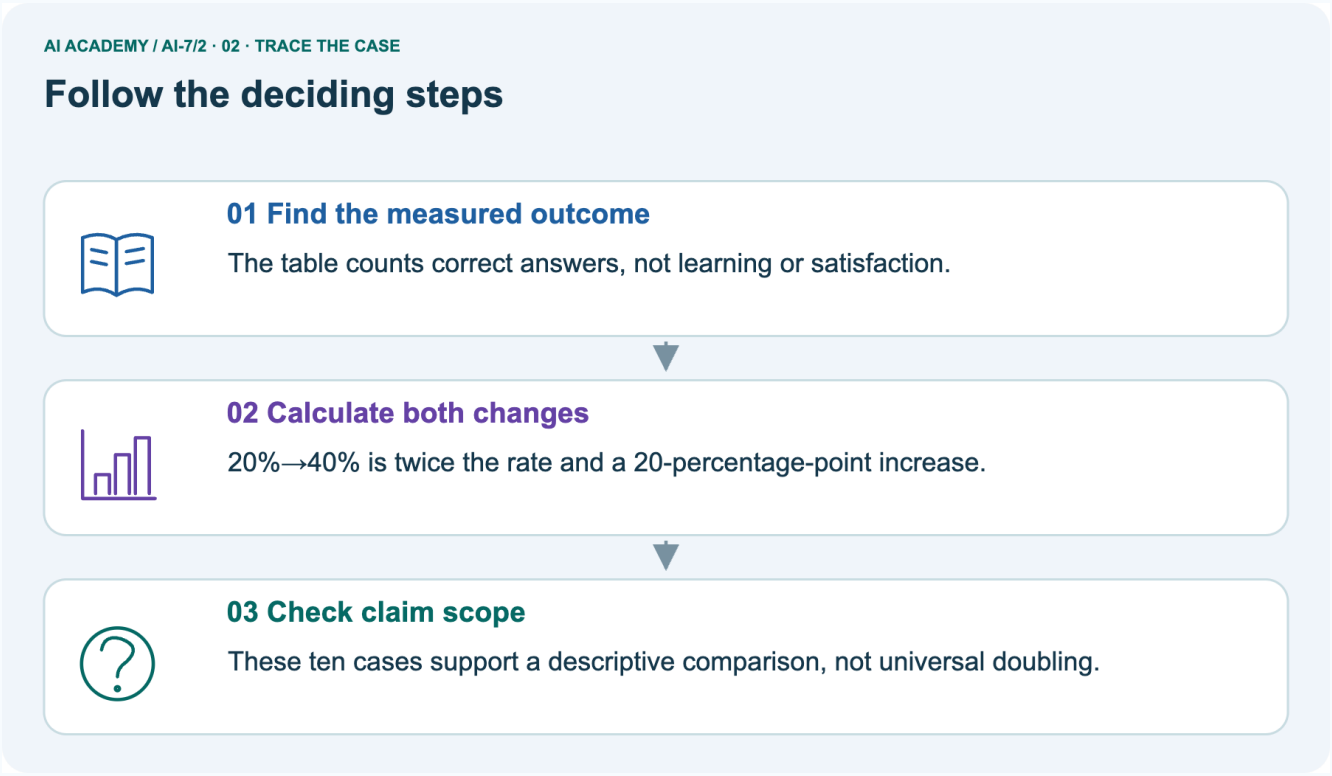
Baseline: 2/10 correct. New prompt: 4/10 on the same ten cases. Scoring is fixed.

The picture is a separate fictional prompt comparison: baseline 2/10 correct versus new 4/10, both under the same ten-case scoring. Rates are 20% and 40%. The latter is twice the former, a 100% relative increase, but the absolute change is 20 percentage points and the new prompt still fails six of ten cases. None of these quantities measures each learner's improvement.

The changed headline "every learner improves twice as much" changes both the unit and outcome without supplying learner data. No significance, population interval or causal study is provided; do not replace that gap with either a claim of statistical significance or a claim that no real effect exists.

Separate picture quantity	Calculation	Meaning and limit
Baseline correctness	2/10 × 100 = 20%	Observed correct-answer rate
New correctness	4/10 × 100 = 40%	Observed rate; six cases still incorrect
Absolute rate change	40% – 20% = 20 percentage points	Not a 20% relative increase
Rate ratio	40% / 20% = 2	Twice the observed rate
Relative increase	(40% – 20%)/20% = 100%	Not twice every learner's improvement
Learner outcome	No learner measurements supplied	Cannot infer individual learning effects

Picture trace and unsupported-learner-headline key



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Find the measured outcome	The table counts correct answers, not learning or satisfaction.
2. Calculate both changes	20%→40% is twice the rate and a 20-percentage-point increase.
3. Check claim scope	These ten cases support a descriptive comparison, not universal doubling.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A research **claim** compresses a method and result. Unpack the measured variable, denominator and comparison before accepting the headline.

Small case counts leave substantial uncertainty; no significance or causal effect is supplied.

Change: The headline changes to "every learner improves twice as much".

Prediction: That learner **claim** is unsupported by these answer counts.

Reason: The outcome and experimental unit have changed in the wording without new **evidence**.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Claim	A statement offered as a conclusion
Evidence scope	The conditions represented by observations
Extrapolation	Extending beyond measured conditions

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

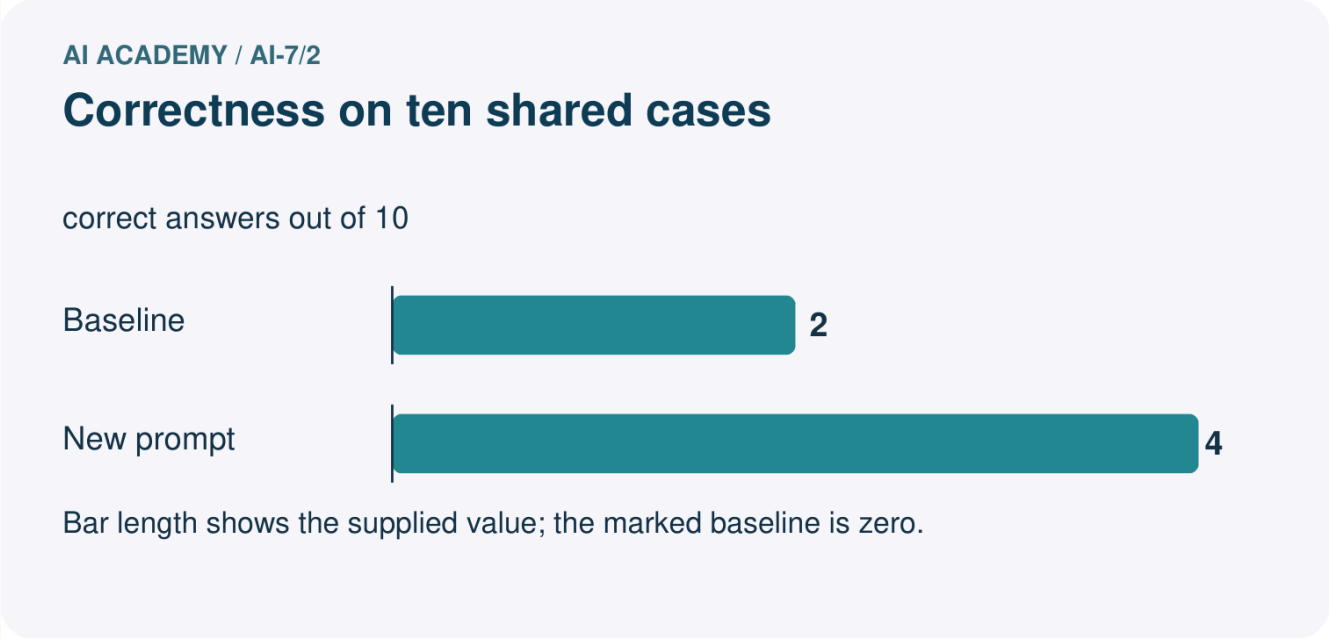
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Reading a research claim

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/2

Reading a research claim

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Find the measured outcome

The table counts correct answers, not learning or satisfaction.



2. Calculate both changes

20% -> 40% is twice the rate and a 20-percentage-point increase.



3. Check claim scope

These ten cases support a descriptive comparison, not universal doubling.

CHANGE ONE THING

The headline changes to "every learner improves twice as much".

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: A fictional report says method B is better after 6/8 supported answers versus baseline 5/8 on the same constructed cases. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: B has 75% versus 62.5%, a 12.5-percentage-point difference on eight cases. Better in this measured outcome does not establish broader tasks or uncertainty.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Reading a research **claim**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: A numerical improvement can be real in the sample while the broader **claim** remains unsupported.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Calculate the baseline and new rates for 2/10 and 4/10. Explain both twice the rate and the percentage-point change, and say why neither measures every learner’s improvement.

The table counts correct answers, not learning or satisfaction. 20%→40% is twice the rate and a 20-percentage-point increase. These ten cases support a descriptive comparison, not universal doubling.

Changed case: That learner **claim** is unsupported by these answer counts. The outcome and experimental unit have changed in the wording without new **evidence**.

Changed-case reasoning: That learner **claim** is unsupported by these answer counts. The outcome and experimental unit have changed in the wording without new **evidence**.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use the worked example in Sections 3 and 6. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain See how it works and Words that unlock the idea. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main observed supported-answer rates 62.5/75% differ by 12.5 percentage points; picture correctness 20/40% doubles relatively but rises 20 points and supplies no individual-learning data. Added paired possibilities are illustrations only. Selected medians 2 versus 3 seconds on four local fixtures support an observed local one-second median difference, not universal speed.

Require workload, environment, timing boundary and quality records; no raw distributions, uncertainty or causal result is provided.

Diagnose the current misconceptions

The worked result establishes every future case: A numerical improvement can be real in the sample while the broader **claim** remains unsupported.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Section 9 for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.

What the response shows	Next teaching action
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new research scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Report **claim**s universal speed after median 2 seconds versus 3 on four local fixtures. Audit scope.

Teacher explanation: Observed median difference 1 second in four local fixtures supports that comparison only; workload, environment and quality must be recorded.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: Reading a **claim** requires its method, data, comparison and uncertainty. Titles and abstracts may omit conditions crucial to interpretation. Separate the **claim** from supplied **evidence**.

Record dataset, unit, baseline, metric and limitations. Ask whether the statement extends beyond those **observation**s.

W2 Trace the worked case

A fictional report says method B is better after 6/8 supported answers versus baseline 5/8 on the same constructed cases. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: B has 75% versus 62.5%, a 12.5-percentage-point difference on eight cases. Better in this measured outcome does not establish broader tasks or uncertainty.

W3 Audit the limit

A report omits this qualification: A numerical improvement can be real in the sample while the broader **claim** remains unsupported.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: A numerical improvement can be real in the sample while the broader **claim** remains unsupported.

State the observed result separately from the unverified or out-of-scope **claim**; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. A numerical improvement can be real in the sample while the broader **claim** remains unsupported.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: B has 75% versus 62.5%, a 12.5-percentage-point difference on eight cases. Better in this measured outcome does not establish broader tasks or uncertainty. Qualification: A numerical improvement can be real in the sample while the broader **claim** remains unsupported. Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a **claim** resting on unmeasured **evidence**.

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

Report **claim**s universal speed after median 2 seconds versus 3 on four local fixtures. Audit scope.

Teacher explanation: Observed median difference 1 second in four local fixtures supports that comparison only; workload, environment and quality must be recorded.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: Observed median difference 1 second in four local fixtures supports that comparison only; workload, environment and quality must be recorded. Require intermediate reasoning and the stated scope. Fresh retry: Report **claim**s zero errors from ten passing fixtures. Expected reasoning: Ten observed passes do not prove zero population or future errors; retain the tested scope. If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit research reasoning

> Use saved evidence and assumptions to choose the next question.

Project and next-week bridge

Add a labelled reading a research **claim evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of reading a research **claim**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Finding primary research. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the **claim** to its records, units and assumptions so another investigator can challenge or reproduce it.

Delayed return and support record

Suggested return points in this level: ai-7/3, ai-7/5, ai-7/8. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a **claim**ed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of reading a research **claim**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

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> **Resources:** Use Find and repair a mistake, Practise together and Run the investigation, with workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

Before class, solve the supported practice: A fictional report says method B is better after 6/8 supported answers versus baseline 5/8 on the same constructed cases. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: B has 75% versus 62.5%, a 12.5-percentage-point difference on eight cases. Better in this measured outcome does not establish broader tasks or uncertainty.

Check every learner before discussion: Use the worked case for Reading a research **claim**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: A numerical improvement can be real in the sample while the broader **claim** remains unsupported.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: The headline changes to "every learner improves twice as much". Require a **prediction** and a reason before comparing. Expected result: That learner **claim** is unsupported by these answer counts.

Reason: The outcome and experimental unit have changed in the wording without new **evidence**.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

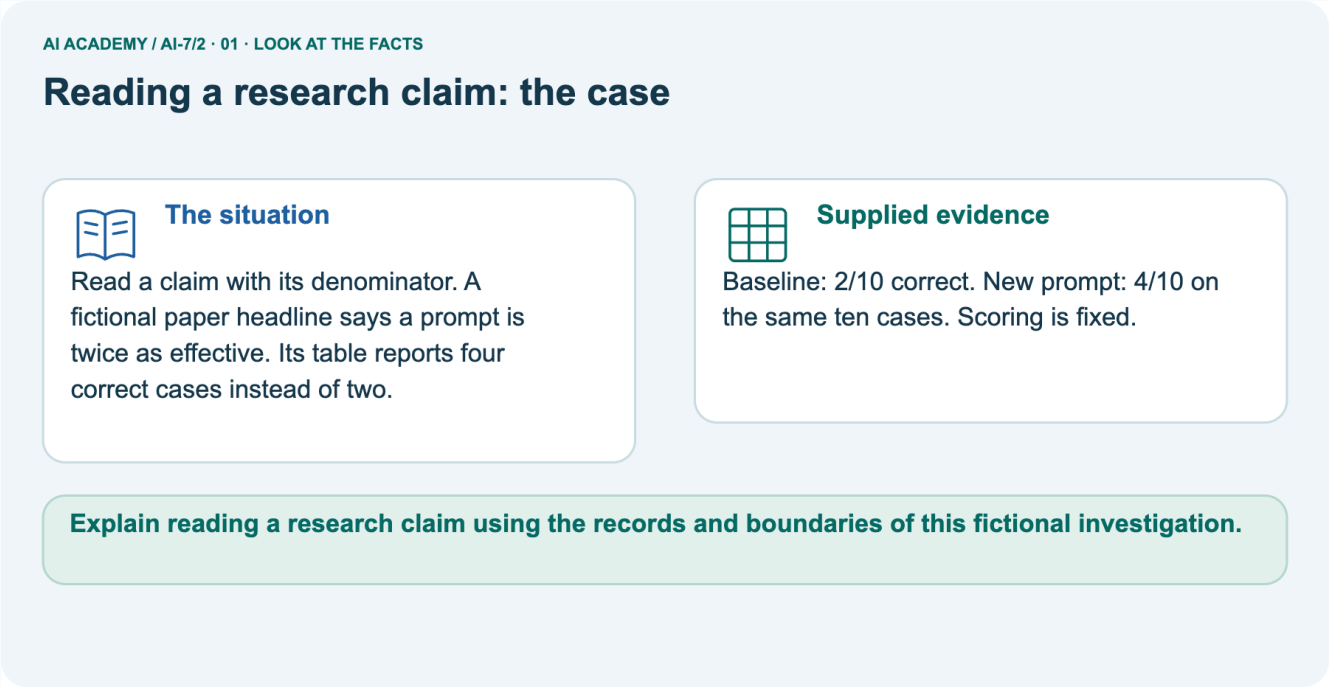
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Reading a research claim: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

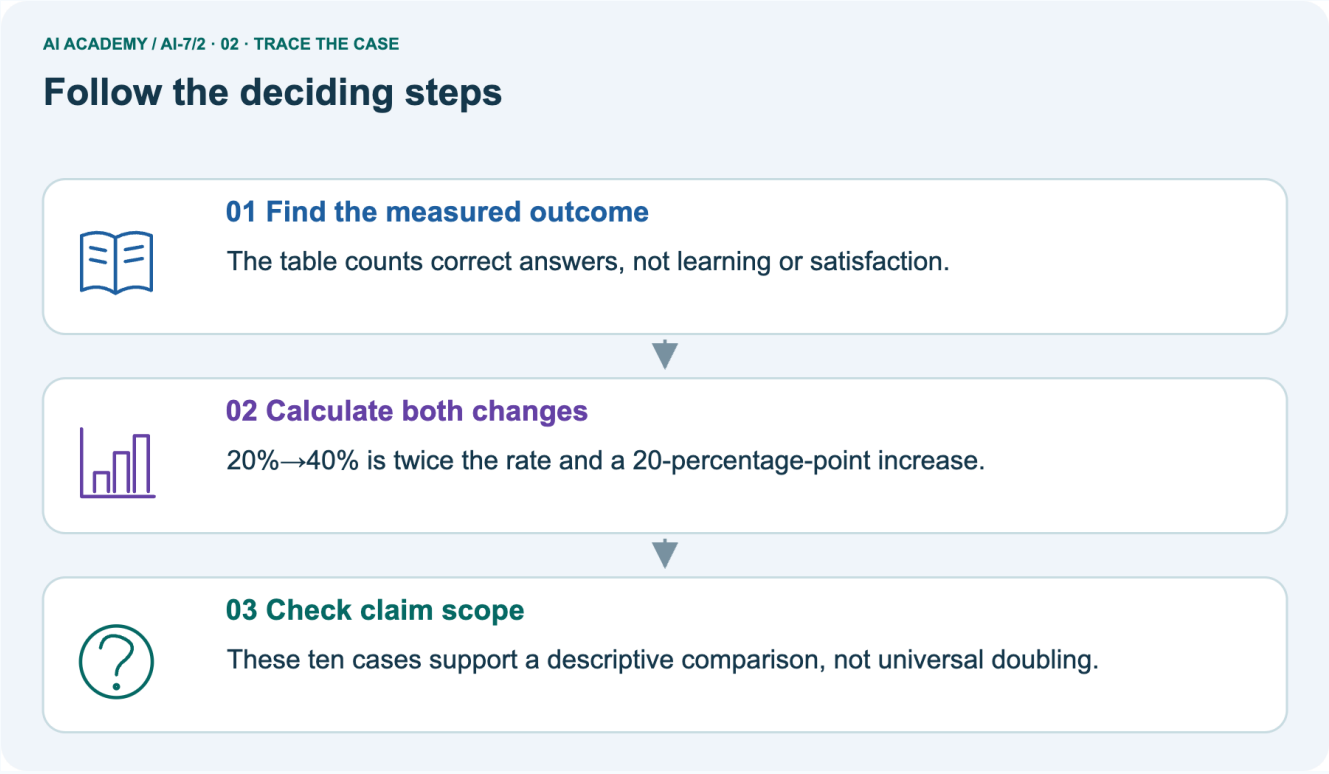
Read a **claim** with its denominator. A fictional paper headline says a prompt is twice as effective. Its table reports four correct cases instead of two. Supplied facts: Baseline: 2/10 correct.

New prompt: 4/10 on the same ten cases. Scoring is fixed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



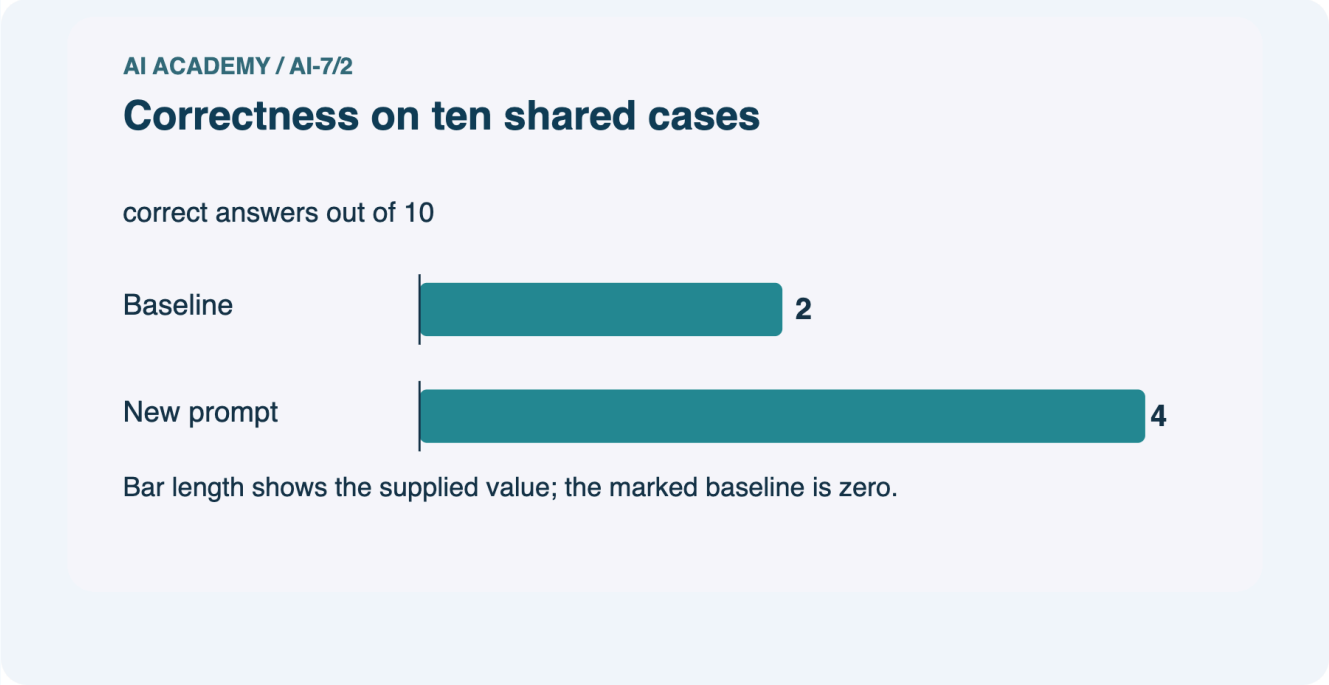
Follow the labelled picture with the matching explanation. Open the vector diagram

Find the measured outcome: The table counts correct answers, not learning or satisfaction.
→ Calculate both changes: 20%→40% is twice the rate and a 20-percentage-point increase.
→ Check **claim** scope: These ten cases support a descriptive comparison, not universal doubling.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Correctness on ten shared cases



Follow the labelled picture with the matching explanation. Open the vector diagram

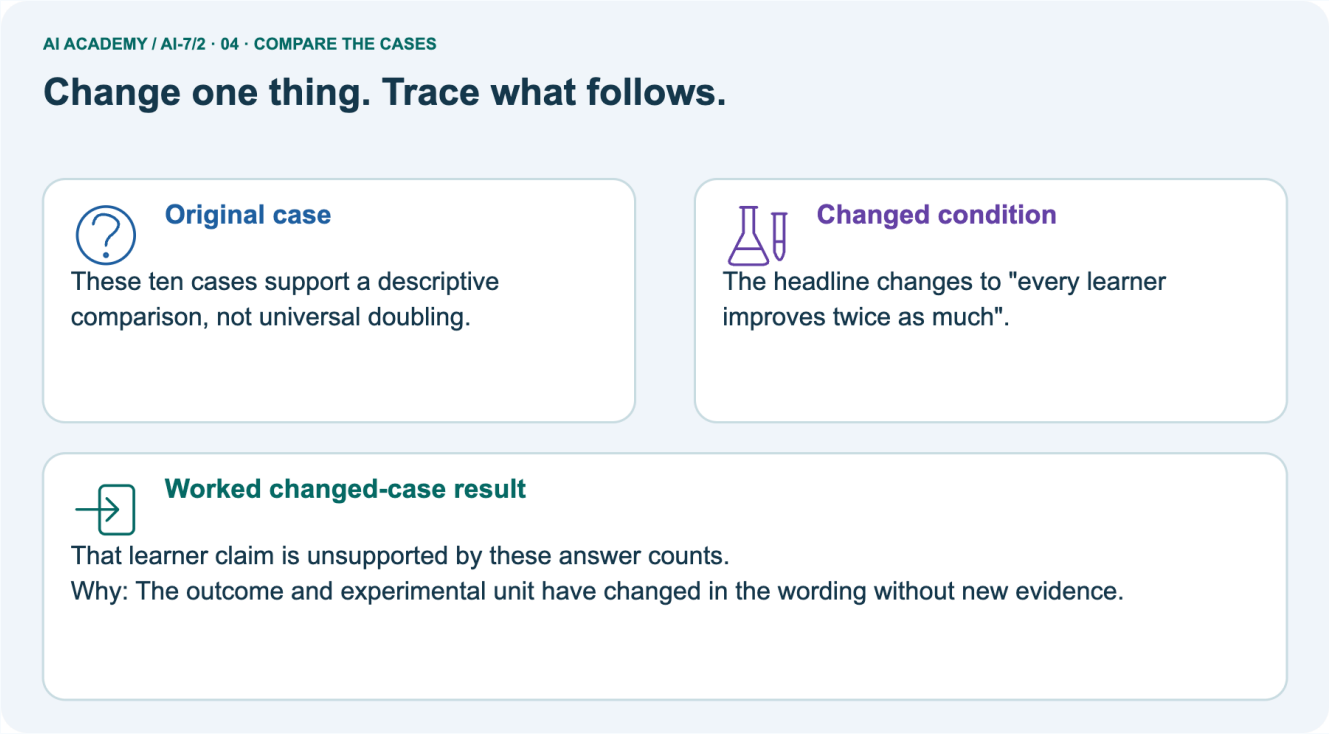
Correctness on ten shared cases. Units: correct answers out of 10. Baseline: 2. New prompt: 4.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: These ten cases support a descriptive comparison, not universal doubling. Changed condition: The headline changes to "every learner improves twice as much".

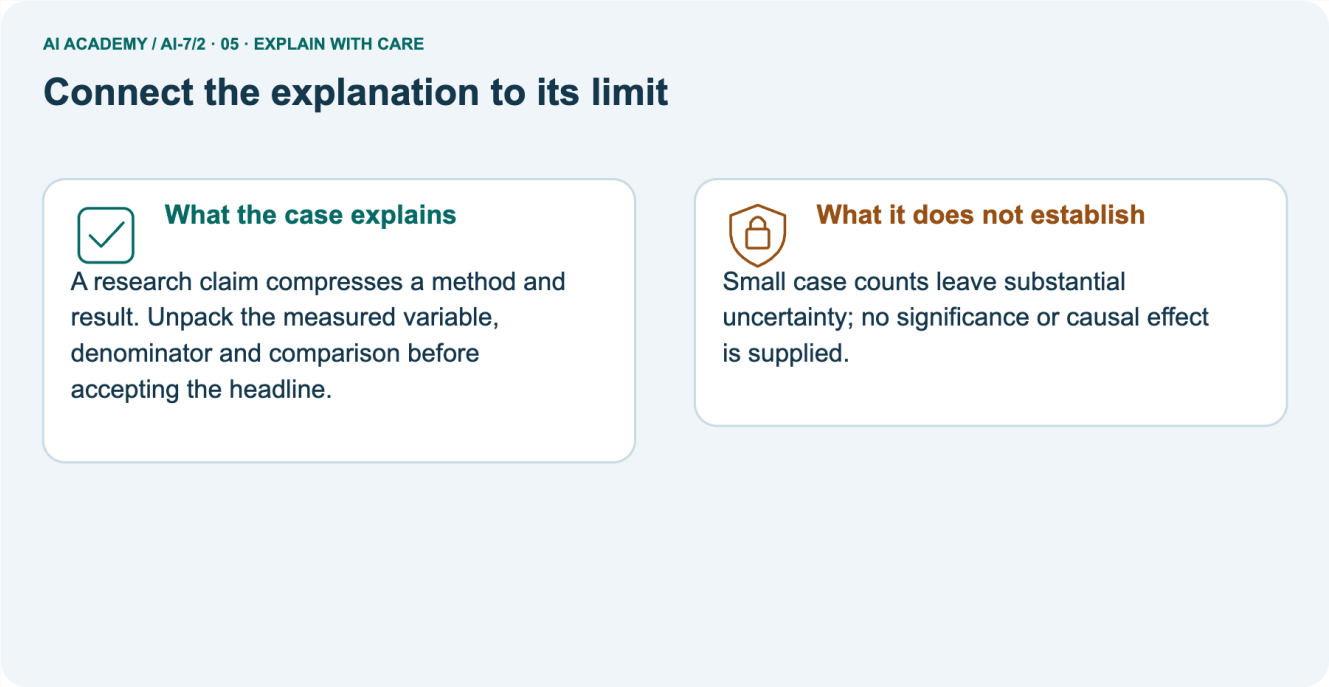
Expected result: That learner **claim** is unsupported by these answer counts. Reason: The outcome and experimental unit have changed in the wording without new **evidence**.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The outcome and experimental unit have changed in the wording without new **evidence**.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A research **claim** compresses a method and result. Unpack the measured variable, denominator and comparison before accepting the headline.

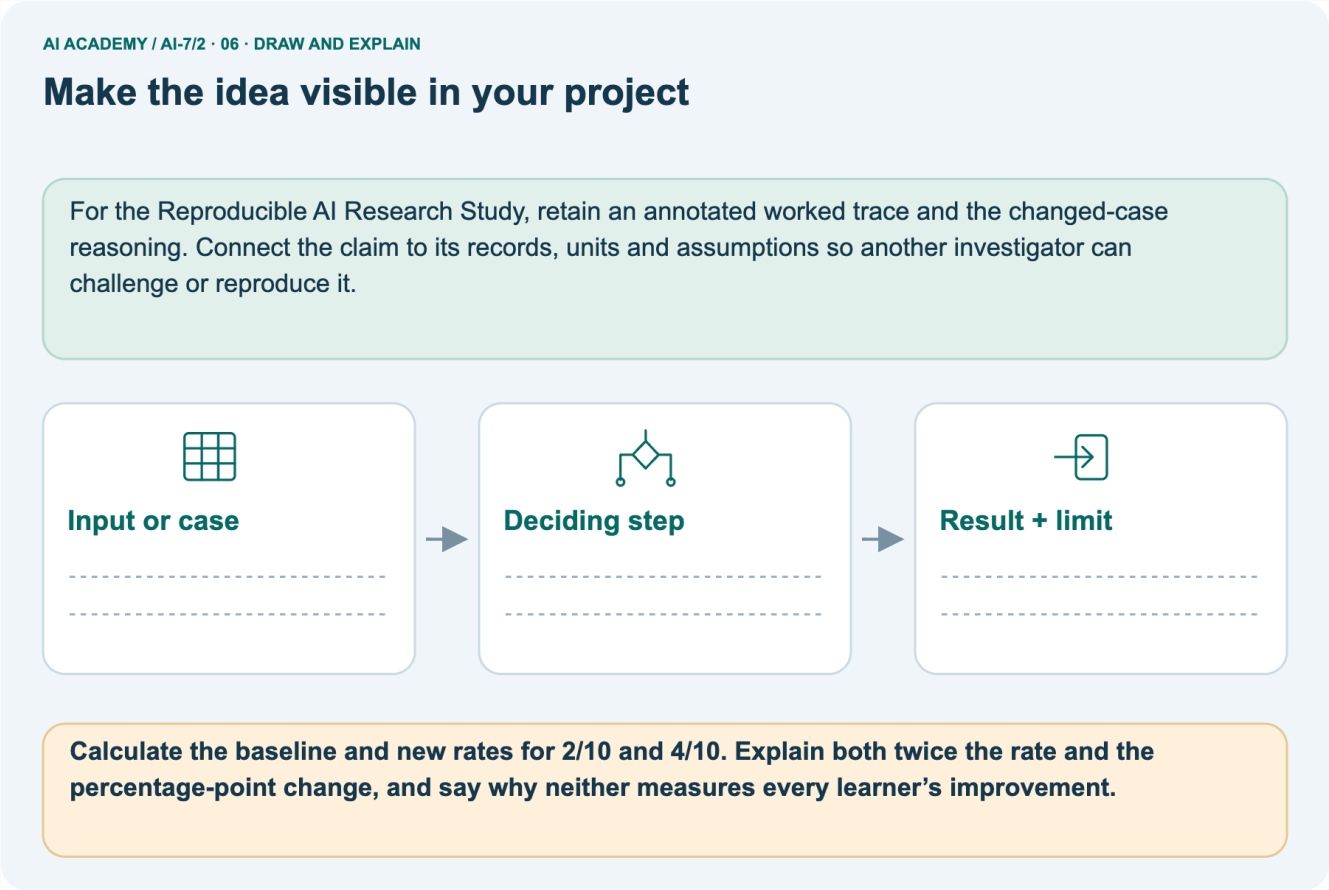
Limit: Small case counts leave substantial uncertainty; no significance or causal effect is supplied.

Explain the case, then point to a limit on the **claim**.

Teacher prompt: Explain the case, then point to a limit on the **claim**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Small case counts leave substantial uncertainty; no significance or causal effect is supplied.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the **claim** to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate the baseline and new rates for 2/10 and 4/10.

Explain both twice the rate and the percentage-point change, and say why neither measures every learner's improvement.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The table counts correct answers, not learning or satisfaction. 20%→40% is twice the rate and a 20-percentage-point increase. These ten cases support a descriptive comparison, not universal doubling.

Changed case: That learner **claim** is unsupported by these answer counts. The outcome and experimental unit have changed in the wording without new **evidence**.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Reading a research claim: the case

02. Follow the deciding steps

03. Correctness on ten shared cases

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.